STAT 425 Statistical Modeling I – Final Project Instruction

Spring 2025

- **Due Date**: Tuesday, May 13th at 11:59 pm
- The final project is a group project, with each group consisting of a maximum of three members. The final team sign up result is available in the module "General Course Information" in Canvas
- Please explore and analyze the Student Habits data (`StudentsHabits.csv` in Canvas). We aim to study how different habits of the students affect their academic performance, as measure by an average exam score. In particular, we ask the following questions:

  What habits affect the academic performance of students significantly? How do they affect the exam score, positively or negatively? How much does a student's exam score increase/decrease on average, if the students changes a habit in a particular manner (e.g., sleep one more hour per day)?

  A more detailed introduction to the data set and some initial R code for exploration is available (`DataExploration.md`).
- Each group needs to submit a final report prepared using R markdown. You will submit both the markdown file and the knitted pdf file with **a maximum length of 15 pages** (including title page, figures, tables and appendix (if any)). **We strongly encourage fewerpages and conciseness in your report**. Please include the following sections:
  - **Title Page**: Title of the report. Group members' names with brief descriptions of each member's contributionsto project development and report preparation.
  - **Abstract**: Briefly outlines the problem, methods, and key findings.
  - **Exploratory Data Analysis**: Focus on meaningful graphical displays and key numerical summaries of the data. Briefly summarize your observations, e.g., what variables are categorical, what variables are highly correlated, are there non-linear patterns, etc.
  - **Model Building**: Include model fitting, diagnostics/remedials, selection of the final model using model comparison techniques, etc.
    * Fit some regression models to answer the main questions above. Constrain yourself to the methods introduced in STAT 425. E.g., linear regression, nonparametric regression, ANCOVA, ANOVA, shrinkage methods. To implement some methods, you will need to perform some model selection or select hyperparameters (e.g., λ in ridge regression).
    * Perform model diagnostics for the models you are interested in. Perform necessary remedials if needed.

    * Choose two or more models that you find worth exploiting. Compare these models using appropriate techniques introduced in STAT 425 (e.g., F-test, AIC, BIC, cross-validation, etc), and select a final model. Some possible model comparisons include
        · A reduced model based on combined variables vs. a full model
        · PCR vs. Lasso vs. Ridge
        · Linear regression vs. nonlinear regression
        · Additive model vs. interaction model
        · ...

  For each step, briefly explain the methods you are using, the findings from your diagnostics, and the reasons behind your final model selection.
  - Discussion of Results and Conclusions: Summarizes findings and insights (can you answer the main question?), discuss open questions or remaining challenges (e.g., some issues you diagnosed but didn't know how to address), etc.

- In the R markdown file, use 'echo = FALSE' for each code chunk to prevent code showing in the knitted pdf.

- The submitted R markdown file should include all the relevant R code for your analysis, and it should be able to knit the pdf file. **Make sure that your R code will run successfully.**

- You are permitted to make multiple submissions. It's important to note that only the last attempt will be graded.

- **Late submission will not be accepted.** To be safe, I recommend you to submit earlier on the due date even if you are unsure of your report; you can always submit a better version before it is due.

- General Rules and Academic Integrity
  - Discussion of the final project with individuals or groups outside your designated working group is strictly prohibited.
  - Utilization of online resources is permitted. However, all sources must be acknowledgedi n an "Acknowledgement" section at the conclusion of the report.
  - Direct copying of sentences or content from the work of others is plagiarismand is strictly prohibited.

Enjoy your statistical modeling exploration!